

flea

NEWS 91 *May 2023*



<http://esanetworks.org/>

FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Before including in publications any of the citations given here, please review for accuracy and correct formatting. Many of our sources are abstracting journals and current literature sources, and editors have not necessarily checked citations.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: New flea species descriptions, Flea research citations from journals or conference proceedings that are not available or indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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John Gay, *The Man and the Flea* (c.1720) with illustration

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Editorial

"*Exploring fleas*"

"I think the age of exploration is just beginning, not ending, on our planet."

- Robert Ballard

"Yet, if we have the capacity to destroy the earth, so too do we have the capacity to protect it."

- Dalai Lama

Dear *Flea News* Reader,

The flea gut is layered complexly. But what is the significance?

The observation has been followed up with research showing energy costs to the flea of digesting blood depend in part on the host involved (Sarfati et al. 2005). What components of the gut or blood are involved that determine the ease of digestibility? Is it possible that digestibility of blood helps determine host specificity of fleas?

The bulk of research conducted on comparative flea anatomy is of hard anatomy (exoskeleton), after a flea is cleared with KOH for mounting on slides. Exoskeletal features have provided much insight into flea phylogeny, and soft anatomy could be helpful additionally for understanding the physiology of sensory, digestive, thermal, and respiratory systems, which in turn are a basis for the physiological ecology of community ecology and evolution of fleas.

Let us now consider biodiversity as a frontier for exploration and protection.

Yours in fleas,
R.L. Bossard
Editor, *Flea News*

Literature

Sarfati, M. et al. 2005. Energy costs of blood digestion in a host-specific haematophagous parasite. *J. Exp. Biol.* 208(13): 2489-2496.

Announcements

Lewis World Flea Species List

Updated 12 February 2022 and available on Flea News network on Entomological Society of America Networks

Search for candidates for doctoral program on fleas

Search for candidates for 5-year doctoral program that will involve studying *Polygenis* fleas (taxonomy and epidemiology), Centro de Estudios Parasit. Vectores, CEPAVE (CONICET-UNLP), Argentina. Contact Dra. Marcela Lareschi (mlareschi@cepave.edu.ar) before 30 June 2023 for more information.

Request for papers for Special Issue on "Taxonomy, Epidemiology, Evolution and Phylogeny of Common Ectoparasites of Vertebrates: Siphonaptera and Acari" of journal *Diversity*

Submission deadline is 15 November 2023 or submitted immediately. Guest Editors are Prof. Dr. Antonio Zurita and Dr. Marcela Lareschi. For more information, visit <https://www.mdpi.com/si/diversity/8KDU234563>.

2023 Electropenetrography (EPG) Online Workshop

"Workshop was designed and recorded by Dr. Elaine A. Backus of the USDA Agricultural Research Service. The Workshop offers instructional videos to learn how to do all aspects of AC-DC EPG (with applicability to DC EPG), plus lectures on the history, principles, types of instruments, and waveforms of major insect groups. There is also a guest lecture on histological correlations by Dr. Holly Shugart of Pennsylvania State University....Register through the UF IFAS Extension online education link."

2024 International Electropenetrography (EPG) Conference

<https://crec.ifas.ufl.edu/extension/epg>. Lake Alfred, FL, USA (near Orlando), January 29-February 2, 2024, Conference registration and abstract submission opens July 1, 2023.

Featured Research

New Flea Species 2022

(none)

Taxonomy

Medvedev, S. G., Verzhutsky, D. B., & Kotti, B. K. (2022). Diversity of plague vectors: Fleas of the genus *Paradoxopsyllus* Miyajima et Koidzumi, 1909 (Siphonaptera, Leptopsyllidae). *Entomological Review*, 102(5), 571-585. doi:10.1134/S0013873822050037

The taxonomic diversity, distribution patterns, and host-parasite relationships of fleas of the Palaearctic genus *Paradoxopsyllus* (Leptopsyllidae, Paradoxopsyllinae) are analyzed. The role of some species of this genus as pathogen vectors and reservoirs in natural plague foci of Eurasia is considered; 8 out of 45 *Paradoxopsyllus* species have been recorded as the main, accessory or occasional vectors of the plague pathogen.

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Zurita, A., Lareschi, M., & Cutillas, C. (2023). New insights into the taxonomy of Malacopsylloidea superfamily (Siphonaptera) based on morphological, molecular and phylogenetic characterization of *Phthiropsylla agenoris* (Malacopsyllidae) and *Polygenis (Polygenis) rimatus* (Rhopalopsyllidae). *Diversity*, 15(2) doi:10.3390/d15020308

From a phylogenetic point of view, the Malacopsyllidae family and the Rhopalopsyllidae [*sic*, Rhopalopsyllidae] family (comprising Parapsyllinae and Rhopalopsyllinae subfamilies) have been traditionally classified within the Malacopsylloidea superfamily, mostly restricted to South America. The phylogenetic relationships and taxonomic status of Malacopsyllidae and Rhopalopsyllidae [*sic*, Rhopalopsyllidae] have never been assessed since no molecular loci of Malacopsyllidae have been sequenced by any authors, and the phylogeny provided so far was not based on any sort of formal quantitative analysis of flea morphology. Based on these precedents, the objective of this study was to carry out a comparative phylogenetic, molecular and morphological study of two different species belonging to each family, *Phthiropsylla agenoris* (Malacopsylla) and *Polygenis (Polygenis) rimatus* (Rhopalopsyllidae, Rhopalopsyllinae). In this study, we demonstrated the usefulness of several morphological features as diagnostic characters to differentiate between *P. (P.) rimatus* and *P. agenoris*. Using molecular and phylogenetic data, we easily discriminated between the two taxa (*P. agenoris* and *P. (P.) rimatus*) by comparing both nuclear and mitochondrial markers. This fact proves the usefulness of ITS2, EF1- α , *cox1*, *cytb* and *cox2* as molecular diagnostic markers to characterize and identify different Siphonaptera taxa. Additionally, the phylogenetic results confirm, for the first time, the monophyly of the Malacopsyllidae family and suggest a clear paraphyletic position of the Parapsyllinae [*sic*, Parapsyllinae] subfamily and, consequently, the Rhopalopsyllidae family.

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Duffus, Natalie E., & Juliano Morimoto. 2022. Current conservation policies in the UK and Ireland overlook endangered insects and are taxonomically biased towards Lepidoptera. *Biological Conservation* 266 (2022) 109464.

Insects provide key ecosystem services for our sustainable future, which rely upon effective conservation policies to protect insect biodiversity. To date, however, we still do not know how effective current conservation policies are for protecting insect biodiversity, opening up the possibility that policies are unfit-for-purpose. Given the considerable debate and public awareness on the potential global decline of insect species, it is important to understand whether or not current policies can protect insect biodiversity. Here, we used IUCN listing of species status and developed a quantitative framework to analyse the potential effectiveness and coverage of current conservation policies pertaining to insect biodiversity in the UK and Ireland. We contrasted this against coverage for a well-known group – mammals – as a benchmark, to find that while the vast majority of the UK mammalian species in the European IUCN red list are directly protected by current policies, insects remain largely unprotected. Moreover, for those insect species that are explicitly protected by current policies, there is a taxonomic bias whereby the majority (>50%) of insect species are Lepidopterans (moths and butterflies), while a minority are Coleopterans (beetles), and none are Hymenopterans (bees, ants, wasps). Similar trends were observed in the UK priority biodiversity lists. Based on our data, we conclude that current biodiversity policies in the UK and Ireland have significant gaps in their protection of insect biodiversity and, there is a taxonomic bias that may skew some conservation efforts toward butterflies and moths. We anticipate that our findings are likely to occur worldwide, highlighting the need for more directive policies to manage and protect insect biodiversity for the sustainability of ecosystem services.

["The exception to this was the Scottish biodiversity list which included a species of Dermaptera (*Labia minor*), which is a detritivore at the edge of its range in Scotland (Burton and Burton, 2002; NBN Atlas, 2021) and a species of Siphonaptera (*Ceratophyllus (Emmareus) fionnus*), which is endemic on the Isle of Rùm (Kwak et al., 2019)."]

Hosts and Distribution

Amarga, Ace Kevin S., Michael W. Hastriter. 2022. Bat Fleas of the Philippines, with New Distribution Records from Bohol and Balabac Islands (Siphonaptera: Ceratophylloidea: Ischnopsyllidae). *Taiwanese Journal of Entomological Studies* 7(2): 14–26.

Ischnopsyllidae is a highly specialized family consisting of species exclusively ectoparasitic on bats. The Philippine bat flea fauna currently comprises two subfamilies, three genera and four species. Our study presents the new distribution records from Bohol Island and Balabac Island, as well as the first overview of the natural history and distribution of known ischnopsyllid fleas in the Philippines.

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Amarga, Ace Kevin S., Michael W. Hastriter. 2022. First Report of the Bat Ectoparasite Fauna on Siargao Island, Philippines. *Taiwanese Journal of Entomological Studies* 7(4): 50–53.

We provide a preliminary account of the bat ectoparasite fauna of Siargao Island, Philippines. Three bat fly species are recorded in this study, representing the genera *Brachytarsina* Macquart,

Eucampsipoda Kolenati, and Megastrebla Maa. There was no published work on Siargao Island bat flies before this study. Thus, this report represents the first published documentation of bat flies in Siargao Island, Philippines.

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Amarga, Ace Kevin S., Michael W. Hastriter. 2023. First record of the flying fox bat fly *Cyclopodia horsfieldi* de Meijere, 1899 (Diptera: Hippoboscoidea: Nycteribiidae) on Ursula Island, Philippines. *Historia naturalis bulgarica* 45 (2): 25–29.

**

Meddour, S., Mlik, R., Dik, B., Hastriter, M. W., & Sekour, M. (2022). Ectoparasites of the common gundi (*Ctenodactylus gundi* Rothmann) from the Aures region, Algeria. *Annals of Parasitology*, 68(3), 519-529.

Ctenodactylus gundi were captured in the southern region of Aures, Algeria from December 2015 through June 2017 to assess their ectoparasite diversity. 63.5% of the gundi had one or more ectoparasites, while 36.5% had none. Eight species of ectoparasites were identified. The most abundant taxa were species of *Neotrombicula* (61.5%), *Trombicula* (10.6%), and *Caenopsylla mira* (4.8%). The mean intensity for arachnids follows: *Neotrombicula* spp. (14.3), *Trombicula* spp. (10.6), and *Rhipicephalus (Boophilus) annulatus* (1.6). Among arachnids, mean intensity of males was greater than females in contrast to that of the flea *C. mira*, whose mean intensity for females was 4.8 compared to that of males with 2.8. *C. mira* and *Xenopsylla cheopis* have previously been reported on the gundi, while the mites *Neotrombicula* spp., *Trombicula* spp., *Eulaelaps stabularis*, *O. bacoti*, the tick *Rh. (Boophilus) annulatus* and the louse *Polyplax serrata* are reported here for the first time. Prevalence was influenced by host sex and was higher in males than females. The highest prevalence was recorded in species of *Neotrombicula* (39.2%) and *Trombicula* (20.3%). The abundance of these ectoparasites was directly influenced with climatic conditions specially with temperature ($P=0.0002$), and air humidity ($P=0.0014$).

[Editor's note: The abstract mentions that gundi have *X. cheopis* (Pulicidae), but the authors found *X. ramesis*, along with *C. mira* (Leptopsyllidae).]

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Oliver, G.V., Eckerlin, R.P. Fleas (Siphonaptera) From the Puma, *Puma concolor* (Carnivora: Felidae), A Rangewide Review and New Records from Utah and Texas, USA (2022) *Journal of medical entomology*, 59 (6), pp. 2045-2052.

Fleas have rarely been reported from the puma, *Puma concolor* (Linnaeus, 1771), despite its vast geographic range, its breadth of habitat use, and its diverse diet, all of which bring it into contact with many other species of mammals and potentially their fleas. We review the reported occurrence of 8 species of fleas from pumas, 7 of these species being from wild hosts and 1 species from a host in captivity, and we correct the mistaken report of 1 other flea species from the puma. We present 10 new records of 4 species of fleas from the puma in Utah and Texas. 2 of these flea species, *Cediopsylla inaequalis inaequalis* (Baker, 1895) and *Odontopsyllus dentatus* (Baker, 1904), represent new host records, and 1 species, *Chaetopsylla setosa* Rothschild, 1906, is a new state record for Utah as well as being 1 of the 2 southernmost known localities for this species. At least 7 of the 9 flea species now known from free-ranging pumas are species that are acquired by pumas from their prey. Pumas may be primary hosts of 2 flea species, but even these fleas may be from prey. Some of the flea species that parasitize pumas transmit sylvatic plague, and, since pumas are highly vagile and are known to become

infected with plague, they may spread the disease through their dispersal of infected fleas. Pumas and their fleas also may be involved in the ecology of several other bacterial zoonoses, which are discussed.

Physiology

Peach, Daniel AH and Benjamin J Matthews. 2023. Sensory mechanisms for the shift from phytophagy to haematophagy in mosquitoes. Current Opinion in Insect Science 52(2023) 100930.

The Culicomorpha are an infraorder of several families of bloodfeeding flies, including mosquitoes (Diptera: Culicidae). Here we discuss the evolutionary origins of blood-feeding within the Culicomorpha and review literature that suggests this behaviour may have evolved from ancestral plant-feeding or a combination of plant-feeding and insect-feeding. Sialomic and life-history evidence suggest that plant-feeding, concurrent or not with insect-feeding, is parsimonious as an ancestral diet for Culicomorpha. We review the chemical parsimony observed between vertebrate headspace odours, floral headspace odours, and honeydew headspace odours, which are behaviourally attractive to many of the Culicomorpha. We also review the sensory and neural mechanisms involved in changes in olfactory attraction and we propose this observed chemical parsimony as a hypothesis for an associative mechanism which may have led to the development of blood-feeding from plantfeeding that is consistent with a ‘path of least resistance’ for the sensory changes necessary to undergo host shifts.

Evolution

Feyereisen, R. (2022). The P450 genes of the cat flea, *Ctenocephalides felis*: A CYPome in flux. Current Research in Insect Science, 2 doi:10.1016/j.cris.2022.100032

The genome of the cat flea, an ectoparasite of major veterinary importance and the first representative of the Siphonaptera, is highly unusual among arthropod genomes in showing a variable size and a very large number of gene duplications (Driscoll et al., 2020). The cat flea is the target of several classes of insecticides, justifying the description of its CYPome, the complement of P450s that are an important family of detoxification enzymes. 103 P450 genes were annotated on the nine chromosomes, with an additional 12 genes on small, extrachromosomal scaffolds. Only 34 genes were found as single sequences, with 47 duplicated two to four-fold. This included duplication of genes that are mostly single copy P450 genes in other arthropods. Large clusters of mitochondrial clan P450s were observed, resulting in a CYP12 bloom within this clan to 34 genes, a number of mitochondrial P450s not seen in other animals so far. The variable geometry of the cat flea CYPome poses a challenge to the study of P450 function in this species, and raises the question of the underlying causes of single copy control versus multicopy licence of P450 genes.

Ecology

KRASNOV, B.R.; LÓPEZ BERRIZBEITIA, M. FERNANDA; SANCHEZ J.P.; DÍAZ, M. MÓNICA; LARESCHI, MARCELA; KHOKHLOVA, IRINA S.; GRABOVSKY, VASILY I. 2023. The species composition of local flea assemblages at a small scale in two South American regions is predominantly driven by niche-based mechanisms. Parasitology Research 122: 571–583. doi.org/10.1007/s00436-022-07759-2.

KRASNOV B.R., KHOKHLOVA I.S., KIEFER M., KIEFER D., LARESCHI M., MATTHEE S., SANCHEZ J.P., SHENBROT G.I., STANKO M., L. VAN DER MESCHT. 2023. Multi-site interaction turnover in flea-mammal networks from four continents; application of zeta diversity

concept and multi-site generalised dissimilarity modelling. *Ecological Entomology*, in press.

SANCHEZ J.P., LÓPEZ BERRIZBEITIA F., EZQUIAGA M.C. 2023. Host specificity of flea parasites of mammals from the Andean Biogeographic Region. *Medical and Veterinary Entomology*, in press.

Urdapilleta M., Galliari, C.A., Navarro Febre, T, & M. Lareschi. 2022. Effect of host and environment related factors on the distribution of the ectoparasites of the montane grass mouse *Akodon montensis* (Cricetidae, Sigmodontinae) in the Atlantic Forest ecoregion in northeastern Argentina. *Revista Mexicana de Biodiversidad* 93 (2022): e933894.

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Fellin, E., & Schulte-Hostedde, A. (2022). Effects of ticks on community assemblages of ectoparasites in deer mice. *Ticks and Tick-Borne Diseases*, 13(1) doi:10.1016/j.ttbdis.2021.101846

Ectoparasites are fundamental to ecosystems, playing a key role in trophic regulation. Fleas, mites, and ticks are common hematophagous ectoparasites that infest shared mammalian hosts. One common host in Ontario, Canada, is the deer mouse (*Peromyscus maniculatus*). As the climate warms and the geographic ranges of blacklegged ticks (*Ixodes scapularis*) and American dog ticks (*Dermacentor variabilis*) expand, their introduction to new ecosystems may alter current ectoparasite communities. At three different sites where exposure to ticks varied (both in terms of tick diversity and abundance), ectoparasite community structures found on deer mouse hosts were examined, focusing on species co-occurrences and habitat partitioning on the host. We predicted that when tick species were prevalent, ticks would dominate the micro-habitat attachment sites often inhabited by other parasites, thereby significantly altering parasite community structure. Our results suggest that blacklegged ticks and American dog ticks could have a positive association with each other, but a negative or random association with other ectoparasite species, even when they do not occupy the same attachment site. Sampling site played a significant role in community assemblages as well, possibly due to the differences in tick exposure. As the ticks' ranges expand and they become more abundant, it is important to understand how their prevalence can potentially alter the dynamics in an ectoparasite community, affecting the transmission of pathogens that may spread within an ecosystem, from one host to another.

Extended abstract: “one flea species (*Orchopeas leucopus*) was identified”, “Results differed in co-occurrences when considering hosts as independent versus hosts constrained by their population source. When hosts were considered independently, blacklegged ticks and American dog ticks were positively co-occurring, and both tick species were negatively co-occurring with *Trombiculidae* sp. and fleas. ...”, “... in our study, when blacklegged ticks and American dog ticks were not present on a host, the prevalence of *O. leucopus* and mites was higher in deer mice. These results are also similar to Lutermann et al.'s (2015) findings, where fleas were more prevalent when ticks were removed and vice versa. This may be due to direct competition for resources. Although fleas and mites parasitize *Peromyscus* mice for long periods of time and are consistently impacting their host's immune response (Mize et al., 2011), ticks require larger blood meals for shorter periods of time (McKenzie, 1998), and because they require greater volumes of a shared resource, they may be

outcompeting other blood-feeding ectoparasites.”

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Katahira, H., Eguchi, Y., Hirose, S., Ohtani, Y., Banzai, A., Ohkubo, Y., & Shimamoto, T. (2022). Spillover and spillback risks of ectoparasites by an invasive squirrel *Callosciurus erythraeus* in Kanto region of Japan. International Journal for Parasitology: Parasites and Wildlife, 19, 1-8. doi:10.1016/j.ijppaw.2022.07.006

Invasive organisms can alter host-parasite relationships in a given ecosystem by spreading exotic parasites and/or becoming a new reservoir for native ones. Since these problems affect management programs of the invasive host organisms, it is necessary to monitor them individually. The Pallas's squirrel *Callosciurus erythraeus* is an invasive arboreal mammal introduced into Japan that threatens to exacerbate ecological and public health problems by spreading native and exotic parasites. However, only limited surveys have been available especially for ectoparasites, using the traditional combing method in which the possibility of oversight is inherent. Here, we evaluated the ectoparasite occurrences in Kanto region of Japan, using the whole-shaving method as an alternative approach. As a result of examining 52 hosts from two invaded districts (Yokohama and Yokosuka), chigger mites (*Leptotrombidium* spp.) and fleas (*Ceratophyllus anisus* and *Ceratophyllus indages indages*) were newly recovered in addition to the previously reported tick (*Haemaphysalis flava*) and exotic lice (*Enderleinellus kumadai* and *Neophaematoponiscus callosciuri* [sic, *Neohaematopinus callosciuri*]). The parasite burdens were higher in Yokosuka and in male host individuals, affecting infracommunity richness and composition. Our findings on the variety of native and exotic ectoparasites, at higher abundances in some cases than previously known, may suggest that both the spillover and spillback risks need to be adjusted upwards.

Pathology and Control

Lefèvre, Thierry, Nicolas Sauvion , Rodrigo P.P. Almeida , Florence Fournet, and Haoues Alout. 2022. The ecological significance of arthropod vectors of plant, animal, and human pathogens. Trends in Parasitology, May 2022, Vol. 38, No. 5.

Vector management is a cornerstone in the fight against vector-borne pathogens. However, the impact on ecosystem functioning of reducing or eliminating arthropod vector populations remains poorly understood. Vectors are members of complex ecological communities, and recent studies suggest that their population suppression alters food web dynamics (bottom-up and top-down trophic cascades), inter- and intraspecific competition, and plant pollination. Other possible overlooked roles are also proposed. In this review, with examples from vectors of plant, animal, and human pathogens, we highlight that, although the ecological roles of most vector species might be redundant with other nonvector species, changes in vector abundance alter biotic interactions and, thus, are unlikely to be neutral in terms of ecosystem functioning.

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Mészáros, T. (2022) Egy újabb bejegyzés a Monacensis Graecus 307a jelzetu kéziratban. Antik Tanulmányok, 66 (2), pp. 269-277.

Codex Monacensis Graecus 307a, copied by twelve hands at the beginning of the 16th century, contains the historical work of Laonikos Chalkokondyles. In addition to the main text, the manuscript also preserved several marginal notes. After the text of Chalkokondyles' Apodeixis, on page 237v of the codex, we can find an entry added by Nikolaos Sophianos, in which he describes different methods against fleas and bugs. Although the text of the recipe was already published by Jeno Darkó previously, his version is inaccurate and contains many errors. The aim of the present paper is to provide a new decipherment of this four-line text with translation and short comments.

Microbial Symbiotes

Occhibove, F., McKeown, N. J., Risley, C., & Ironside, J. E. (2022). Eco-epidemiological screening of multi-host wild rodent communities in the UK reveals pathogen strains of zoonotic interest. *International Journal for Parasitology: Parasites and Wildlife*, 17, 278-287. doi:10.1016/j.ijppaw.2022.02.010

Wild rodent communities represent ideal systems to study pathogens and parasites shared among sympatric species. Such studies are useful in the investigation of eco-epidemiological dynamics, improving disease management strategies and reducing zoonotic risk. The aim of this study was to investigate pathogen and parasites shared among rodent species (multi-host community) in West Wales in an area where human/wildlife disease risk was not previously assessed. West Wales is predominantly rural, with human settlements located alongside to grazing areas and semi-natural landscapes, creating a critical human-livestock-wildlife interface. Grounddwelling wild rodent communities in Wales were live-trapped and biological samples – faeces and ectoparasites – collected and screened for a suite of pathogens and parasites that differ in types of transmission and ecology. Faecal samples were examined to detect *Herpesvirus*, *Escherichia coli*, and *Mycobacterium microti*. Ticks and fleas were collected, identified to species based on morphology and genetic barcodes, and then screened for *Anaplasma phagocytophilum*, *Babesia microti*, *Borrelia burgdorferi* sensu lato, and *Bartonella* sp. All the pathogens and parasites screened pose a characteristic epidemiological challenge, such as variable level of generalism, unknown zoonotic potential, and lack of data. The results showed that the bank vole *Myodes glareolus* had the highest prevalence of all pathogens and parasites. Higher flea species diversity was detected than in previous studies, and at least two *Bartonella* species were found circulating, one of which has not previously been detected in the UK. These key findings offer new insights into the distribution of selected pathogen and parasites and subsequent zoonotic risk, and provide new baselines and perspectives for further eco-epidemiological research.

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Špitalská, E., Minichová, L., Hamšíková, Z., Stanko, M., & Kazimírová, M. (2022). *Bartonella*, *Rickettsia*, *Babesia*, and *Hepatozoon* species in fleas (Siphonaptera) infesting small mammals of Slovakia (central Europe). *Pathogens*, 11(8) doi:10.3390/pathogens11080886

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**

Fleas in Art and History

John Gay (b. 1685, Barnstaple, England - d. 1732, London)

From Poetical Works of John Gay, With a Memoir, Vol. 1.

Little, Brown and Company, Boston, Massachusetts, January 1, 1879.

With an illustration.

FABLES. PART I. THE MAN AND THE FLEA. THE MAN AND THE PLEA.

WHETHER on earth, in air, or main,
Sure ev'ry thing alive is vain!
Does not the hawk all fowls survey,
As destin'd only for his prey?
And do not tyrants, prouder things,
Think men were born for slaves to kings?
When the crab views the pearly strands,
Of Tagus, bright with golden sands,
Or crawls beside the coral grove,
And hears the ocean roll above,
"Nature is too profuse," (says he),
"Who gave all these to pleasure me!"
When bord'ring pinks and roses bloom,
And ev'ry garden breathes perfume,
When peaches glow with sunny dyes
Like Laura's cheek when blushes rise,

When with huge figs the branches bend,
 When clusters from the vine depend,
 The snail looks round on flow'r and tree,
 And cries, "All these were made for me!"
 "What dignity's in human nature,"
 Says Man, the most conceited creature,
 As from a cliff he cast his eye,
 And view'd the sea and arched sky!
 The sun was sunk beneath the main,
 The moon and all the starry train
 Hung tbe vast vault of heav'n. The Man
 His contemplation thus began:
 "When I behold this glorious show.
 And the wide watry world below,
 The scaly people of the main,
 The beasts that range the wood or plain,
 The wing'd inhabitants o'f air,
 The day, the night, the various year,
 And know all these by Heav'n design'd
 As gifts to pleasure human-kind,
 I cannot raise my worth too high;
 Of what vast consequence am I!"
 "Not of th'importance you suppose,"
 Replies a Flea upon his nose;
 "Be humble; learn thyself to scan;
 Know, pride was never made for Man.
 'Tis vanity that swells thy mind.
 What, Heav'n and earth for thee design'd!
 For thee! made only for our need,
 That more important Fleas might feed."



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